

Muhammad Shan Tariq

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PRINCIPAL INTERESTS

Partial Differential Equations, Numerical Analysis, Machine Learning

DIGITAL SKILLS

- **Hard Skills:**
 - **Programming Languages:** Swift (iOS App Development), C++, Python, Matlab, LaTeX (Overleaf): Advanced user, HTML, CSS
 - **iOS Development:** RESTful APIs, Firebase (including Local and Push Notifications), Core Data, CocoaPods, Alamofire, UI/UX Design Principles
 - **Tools:** Git (Version Control)
- **Soft Skills:** Teamwork and Collaboration, Adaptability, Communication Skills, Time Management
- **Language Skills:** Urdu, Punjabi, English

EDUCATION

Doctor of Philosophy, Mathematics

Expected September 2029 | Dartmouth, MA

University of Massachusetts Dartmouth, College of Engineering

- **Supervisors:** Dr. Bo Dong ✉ (Primary Supervisor), Dr. Yanlai Chen ✉ (Co-Advisor)
- **Relevant Course Work:** High Performance Scientific Computing, Computational Methods, Numerical Linear Algebra, Numerical Methods for Partial Differential Equations, Mathematics of Deep Learning
- **Research focus:** Numerical solutions of KdV-like systems using the Discontinuous Galerkin method.
- **Cumulative GPA:** 4.00
- **Academic Honor:** Research Assistant in the department of Mathematics.

Bachelor of Science, Physics

Mar 2022 | Islamabad, Pakistan

COMSATS University Islamabad

- Minor in Mathematics
- **Cumulative GPA:** 3.45
- Awarded a full scholarship for undergraduate studies by the Higher Education Commission of Pakistan (HEC-STFS). Graduated 5/104 students at the conclusion of the undergraduate program.
- **Dissertation title:** "Solutions of Non-linear Partial Differential Equations using the Bäcklund Transformation".
 - **Advisor and Co-advisor:** Dr. Waqas Masood ✉ and Dr. Mohsin Siddiq ✉
 - Trained in solving a variety of nonlinear PDEs using the Bäcklund transformation.
 - Trained in finding the N-soliton and N-shock wave solutions of nonlinear PDEs using the "Theorem of Permutability" or the "Bianchi Theorem".

EXPERIENCE

University of Massachusetts Dartmouth, Dartmouth, MA

Aug 2024 – Present

Research Assistant

North Dartmouth, MA

- Developed a Discontinuous Galerkin (DG) scheme for the Korteweg–de Vries (KdV) equation using an implicit penalization technique, improving numerical stability and conservation properties.
- Redesigned an optimized DG formulation that reduces computational cost by requiring the solution of a nonlinear system only at the half time step and a linear system at the next time step, streamlining the process compared to solving nonlinear systems at both levels in the previous approach.

- Developed a DG scheme that enables the implementation of higher-order time-stepping methods (e.g., IRK4), overcoming the limitation of earlier DG scheme that supported only up to second-order methods such as IRK2.

Content Arcade (Pvt.) Ltd.

Mar 2024 – Jun 2024 | Faisalabad, Pakistan

iOS Developer

- Developed and enhanced features for the *Add Text to Photos* app, enabling users to edit images by adding customizable text with various styles.
- Improved UI/UX for a seamless and intuitive photo customization experience on iOS devices.
- Integrated AWS cloud services for efficient data storage and retrieval through API calls.

Research Internship

Feb 2022 – Dec 2022 | Islamabad, Pakistan

National Center for Physics

- Research internship of almost ten months in the Theoretical Physics Department (TPD) for the publication of my research work.
- The scope of the research thesis was expanded to resolve the nonlinear Partial Differential Equation known as the Cylindrical Korteweg-de Vries (CKdV) equation. This was achieved using both the Bäcklund and similarity transformations. Insights into the fluid model within the context of plasmas were gained during an internship, leading to the discovery of N-soliton solutions for the CKdV equation.

Teaching Assistant

Feb 2022 – Dec 2022 | Islamabad, Pakistan

Comsats University

- Following graduation, the research thesis was extended for publication while serving as a Teaching Assistant under the guidance of Dr. Waqas Masood. [↗](#)
- The role specifically involved assisting in the subject "Mathematical Methods for Physics."

Summer Internship

Jun 2021 – Aug 2021 | Islamabad, Pakistan

National Center for Physics

- Worked in the Theoretical Physics Department under the supervision of Dr. Mohsin Siddiq [↗](#) and Dr. Waqas Masood. [↗](#)
- During this internship, learned different mathematical methods to solve linear and nonlinear PDEs.

PEER REVIEWED PUBLICATIONS

M. Shan Tairq, W. Masood, M. Siddiq, S. Asghar, B. M. Alotaibi, Sherif ME Ismaeel, and S. A. El- Tantawy. "Bäcklund transformation for analyzing a cylindrical Korteweg–de Vries equation and investigating multiple soliton solutions in a plasma." *Physics of Fluids* 35, no. 10 (2023). [↗](#)

M. Shan Tariq, Masood, W., Alhejaili, W. *et al.* Investigation of Nonlinear Cylindrical Electrostatic Excitations in Dense Quantum Astrophysical Plasmas. *Braz J Phys* 55, 32 (2025). [↗](#)

ONLINE ADDITIONAL COURSES

- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning
- Advanced Learning Algorithms
- Introduction to iOS Mobile Application Development
- Version Control
- Introduction to Programming with MATLAB
- Python Data Structures
- Programming for Everybody, Getting Started with Python
- Data Visualization using Plotly
- From the Big Bang to Dark Energy
- Introduction to Particle Accelerators (NPAP MOOC)
- Introduction to Quantum Information

AWARDS AND DISTINCTIONS

Undergrad Scholarship

Feb 2018 – Mar 2022 | Islamabad, Pakistan

Comsats University Islamabad

- Recipient of Science Talent Farming Scheme and Higher Education Commission of Pakistan scholarship for Bachelor in Physics. 

Undergrad Position

Nov 2021 | Islamabad, Pakistan

- Secured 3rd position in the 7th semester of undergraduate degree.

Best Ambassador

Mar 2020 | Topi Swabi, Pakistan

- Received the "Best Ambassador" award from the Photo-Optical Instrumentation Engineers and Optical Society of America 

High School Scholarship

Aug 2017 – Jun 2018 | Sahiwal, Pakistan

Government Postgraduate College

- Recipient of the Pakistan Science Foundation  scholarship for two consecutive years during my high school education.

Summer Camp

Jul 2016 | Islamabad, Pakistan

Science Talent Farming scheme, Pakistan Science Foundation

- In my higher secondary education, I successfully cleared three national-level tests in Pakistan, which granted me the privilege of attending a week-long summer camp organized by the Pakistan Science Foundation. 

Scholarship

Oct 2016 | Sahiwal, Pakistan

- Recipient of Punjab Education Endowment Fund  scholarship.

CONFERENCES AND SEMINARS

Pakistan International Conference on Quantum

2021 | Islamabad, Pakistan

Information and Computation (PICQIC-21)

National Center for Physics

- At this conference, we had the opportunity to gain insights into quantum technologies and engage in meaningful discussions with leading experts in the field.

Light Fidelity in Wireless Communication

2019 | Islamabad, Pakistan

Comsats University

- Presented "Light Fidelity in Wireless Communication" in a departmental seminar at Comsats University.

POSTER PRESENTATIONS

Self Assembled Nano-structures

2021 | Islamabad, Pakistan

Air University

- I presented how atoms/molecules/nanoscale building blocks spontaneously organize into ordered structures or patterns with nanometer features without human intervention. I also talked about why these structures are the most promising practical low-cost and high-throughput approaches for nanofabrication.

Photonic Thrusters: The Future of Propulsion System

2020 | Islamabad, Pakistan

Bahria University

- The poster was focused on how the use of laser (photonic thrusters) would work to propel the sails. And how these thrusters would be very useful for us to explore the universe.